

Ibex GP Progress (Zero-RISC)

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Outlines

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- Automated Stubs PDN
- Inquiries

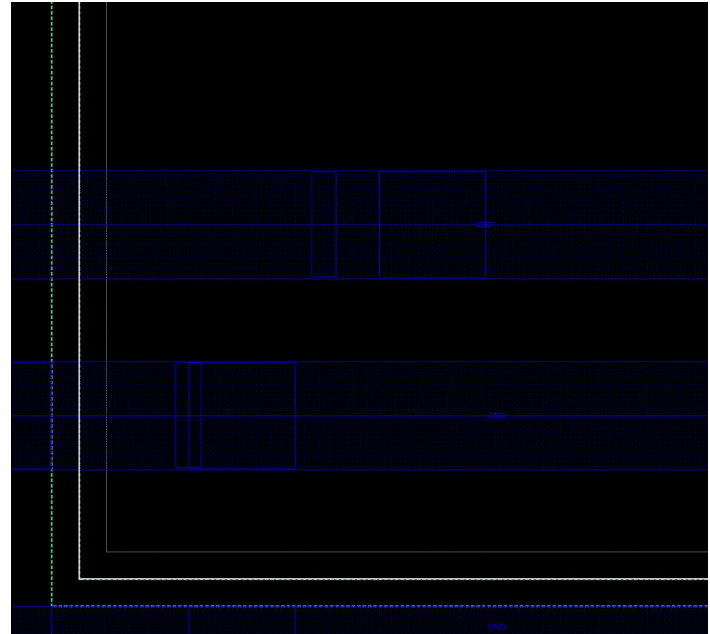
Manual Stubs PDN

Approach

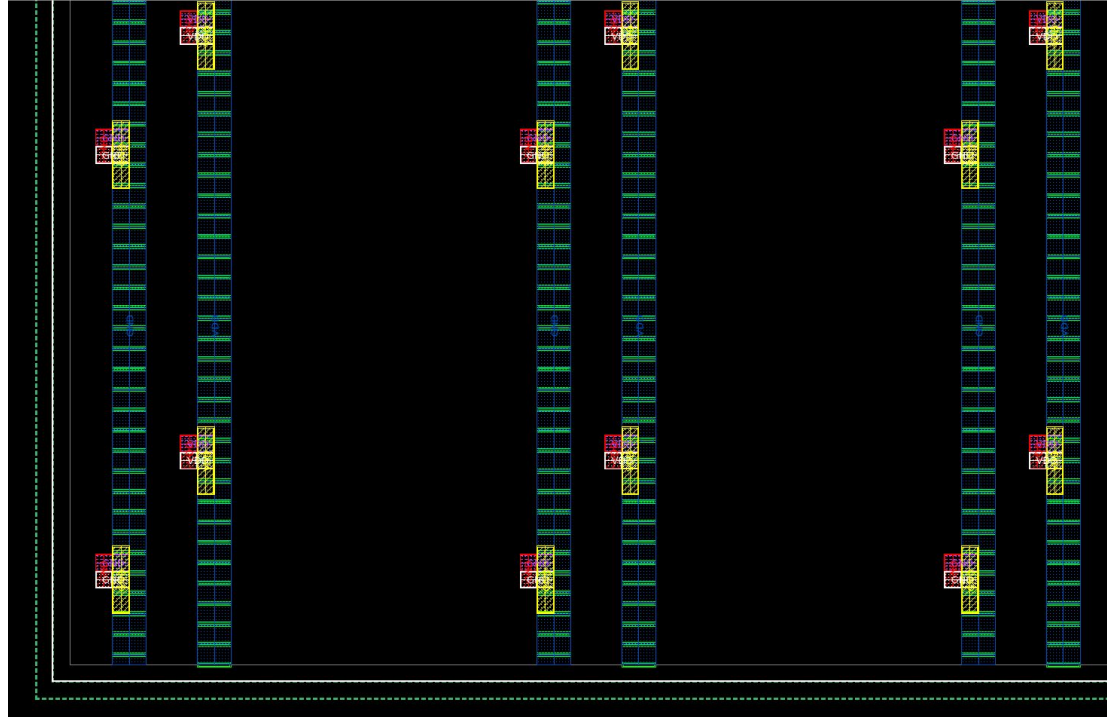
To create stubs orthogonal and partially overlapped to behave as physical building stairs.

The stair stubs are from M7 to M4 then, M3 has vertical continuous stripes to cover all the horizontal cell rails of M1.

Finally, M2 has thin stubs to distribute power from M3 to M1 rails.



Final Look



Script Snippet

```
set_pg_strategy_via_rule adjacent_only \  
-via_rule {{intersection: adjacent}}{via_master: default}}
```

A strategy to guide the tool to place vias only on the intersection between 2 adjacent orthogonal wires.

Script Snippet

```
1 create_pg_wire_pattern SEGMENT \  
2     -parameters {seg_layer seg_direction seg_width seg_low_end seg_high_end} \  
3     -layer @seg_layer \  
4     -direction @seg_direction \  
5     -width @seg_width \  
6     -low_end_reference_point @seg_low_end \  
7     -high_end_reference_point @seg_high_end
```

A wire segment that can be used later with `create_pg_composite_pattern` to distribute that segment over a specified region.

Script Snippet

```
##### M7 #####  
create_pg_composite_pattern M7_PATTERN \  
-parameters {pat_pitch pat_offset_x pat_offset_y} \  
-add_patterns {{name: SEGMENT} \  
               {pitch: @pat_pitch} \  
               {offset: @pat_offset_x @pat_offset_y} \  
               {parameters: {M7 vertical 1 0 4}}}  
  
set_pg_strategy M7_STRATEGY_VDD \  
-core \  
-pattern {{name: M7_PATTERN} \  
          {nets: VDD} \  
          {parameters: {25 8 10}}}  
  
compile_pg -strategies "M7_STRATEGY_VDD" -via_rule adjacent_only
```

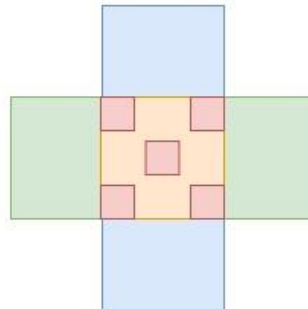
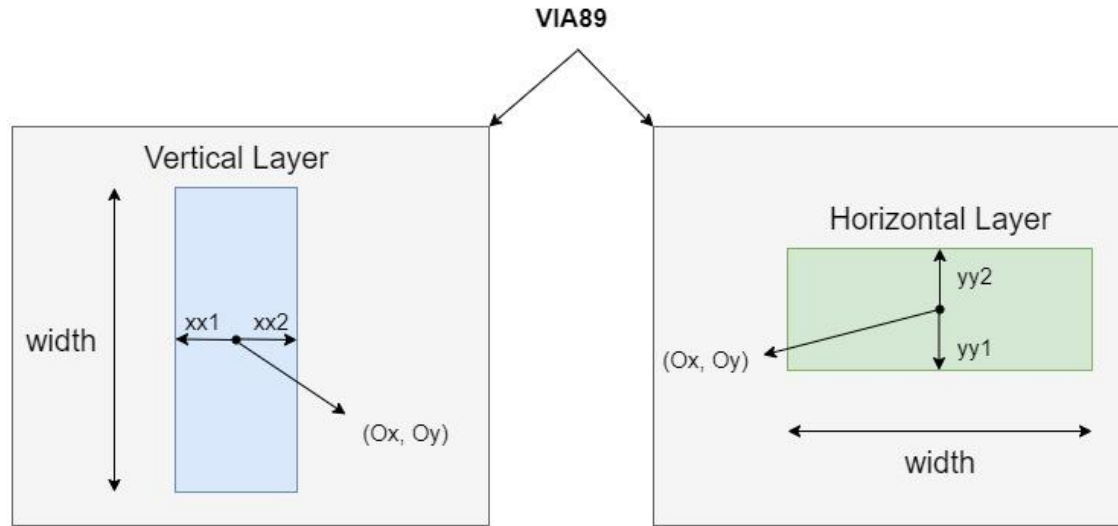
A use example of the segment to draw M7 stubs.

Script Snippet

```
5 create_pg_vias -nets "GND VDD" -from_layers M2 -to_layers M1 -allow_parallel_objects
```

A hidden useful option, `-allow_parallel_object`, was used to place vias from M2 to M1. Both of them are horizontal and their intersection wasn't captured by the tool since the tool defines intersecting as the common area between orthogonal layers only.

Automated Stubs PDN



Automated Stub Script

```
set via_list_VDD [get_vias -within {{18.0000 18.0000} {164.8200 164.8000}} -filter "net_type == power"]

foreach_in_collection v1 $via_list_VDD {
    set orig [get_attribute [get_vias $v1] origin]
    set origx1 [lindex $orig 0]
    set origy1 [lindex $orig 1]
    set xx1 [expr $origx1 - 0.5]
    set xx2 [expr $origx1 + 0.5]
    set yy1 [expr $origy1 - 0.5]
    set yy2 [expr $origy1 + 0.5]

    # creating shape in horizontal layers
    foreach even $MH {
        create_shape -shape_type path -shape_use strip -layer "$even" -path "${origx1 $yy1} {origx1 $yy2}" -net "VDD" -width 2
    }

    # creating shape in vertical layers
    foreach odd $MV {
        create_shape -shape_type path -shape_use strip -layer "$odd" -path "${xx1 $origy1} {xx2 $origy1}" -net "VDD" -width 2
    }

    # creating shape in M3
    create_shape -shape_type path -shape_use strip -layer "M3" -path "${xx1 91.4} {xx2 91.4}" -net "VDD" -width 142.8

    # creating shape in M7
    create_shape -shape_type path -shape_use strip -layer "M7" -path "${xx1 $origy1} {xx2 $origy1}" -net "VDD" -width 4|
}

# creating shape in M2
for {set i 0} {$i < 6} {incr i} {
    set x [expr 32 + [expr 25*$i]]

    for {set j 0} {$j < 120} {incr j} {
        set y1 [expr [expr 20 + [expr 1.2*$j]] - 0.125]
        set y2 [expr [expr 20 + [expr 1.2*$j]] + 0.125]

        create_shape -shape_type path -shape_use strip -layer "M2" -path "${x $y1} {x $y2}" -net "VDD" -width 2
    }
}
```

Automated Stub Script

```
set via_list_GND [get_vias -within {{18.0000 18.0000} {164.8200 164.8000}} -filter "net_type == ground"]

foreach_in_collection v1 $via_list_GND {
    set orig [get_attribute [get_vias $v1] origin]
    set origx1 [lindex $orig 0]
    set origy1 [lindex $orig 1]
    set xx1 [expr $origx1 - 0.5]
    set xx2 [expr $origx1 + 0.5]
    set yy1 [expr $origy1 - 0.5]
    set yy2 [expr $origy1 + 0.5]

    # creating shape in horizontal layers
    foreach even $MH {
        create_shape -shape_type path -shape_use strip -layer "$even" -path "${origx1 $yy1} ${origx1 $yy2}" -net "GND" -width 2
    }

    # creating shape in vertical layers
    foreach odd $MV {
        create_shape -shape_type path -shape_use strip -layer "$odd" -path "${xx1 $origy1} ${xx2 $origy1}" -net "GND" -width 2
    }

    # creating shape in M3
    create_shape -shape_type path -shape_use strip -layer "M3" -path "${xx1 91.4} ${xx2 91.4}" -net "GND" -width 142.8

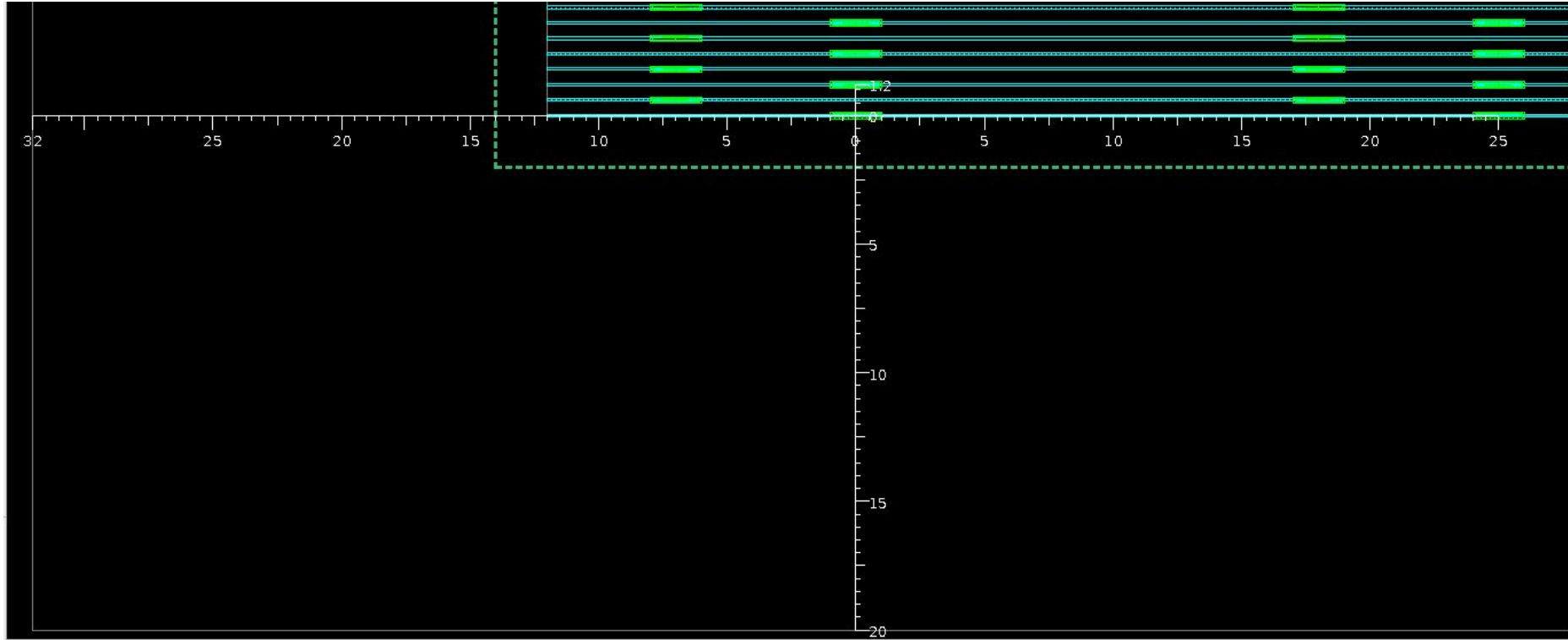
    # creating shape in M7
    create_shape -shape_type path -shape_use strip -layer "M7" -path "${xx1 $origy1} ${xx2 $origy1}" -net "GND" -width 4|
}

# creating shape in M2
for {set i 0} {$i < 6} {incr i} {
    set x [expr 25 + [expr 25*$i]]

    for {set j 0} {$j < 119} {incr j} {
        set y1 [expr [expr 20.6 + [expr 1.2*$j]] - 0.125]
        set y2 [expr [expr 20.6 + [expr 1.2*$j]] + 0.125]

        create_shape -shape_type path -shape_use strip -layer "M2" -path "${x $y1} ${x $y2}" -net "GND" -width 2
    }
}
```

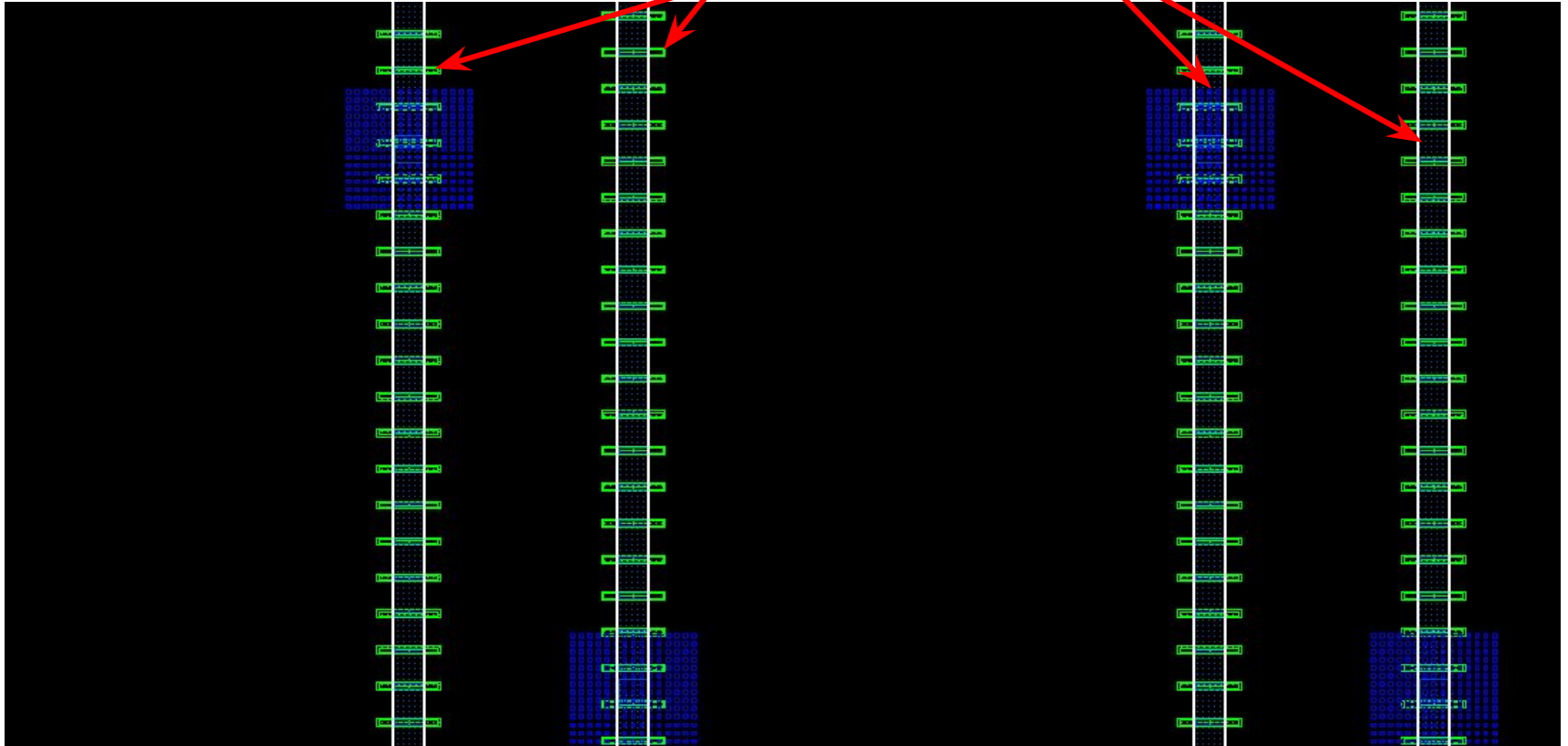
M2 vias



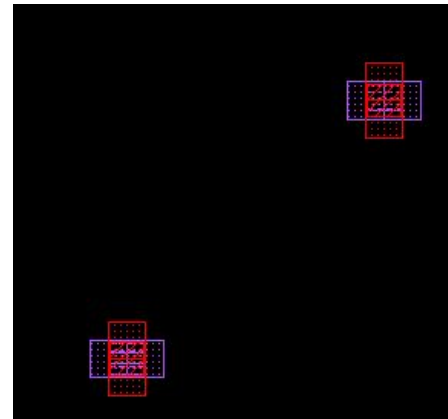
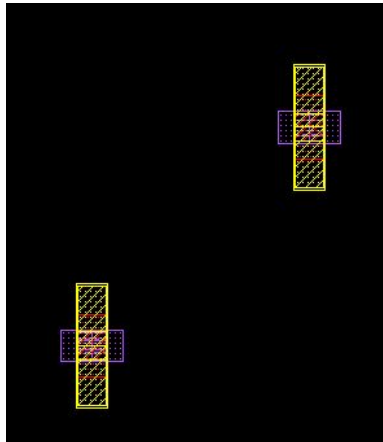
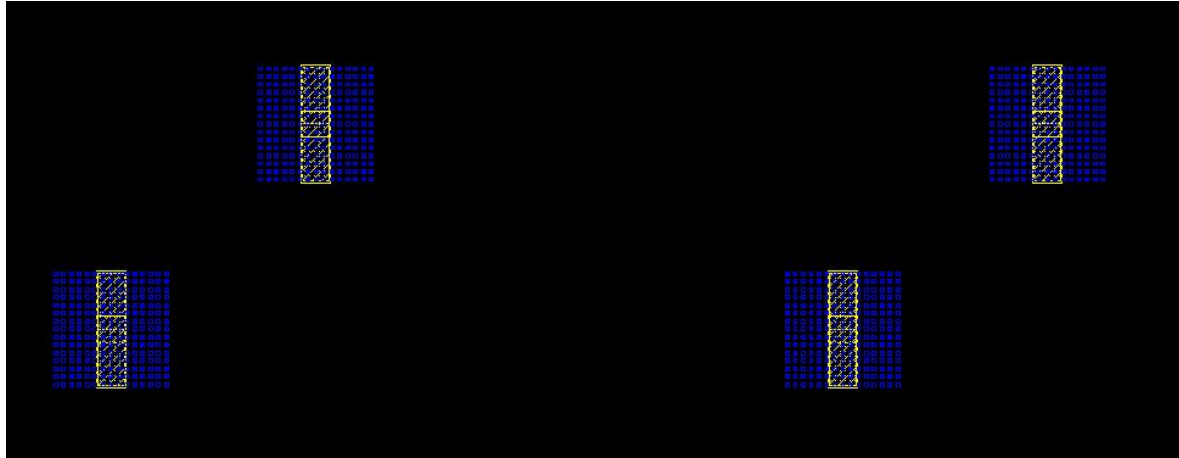
Shape of M2 and M3

M2

M3



Different Width of Metals



Reports

pg_drc

```
1 Command check_pg_drc started at Thu May 22 16:30:35 2025
2 Command check_pg_drc finished at Thu May 22 16:30:35 2025
3 CPU usage for check_pg_drc: 0.20 seconds ( 0.00 hours)
4 Elapsed time for check_pg_drc: 0.20 seconds ( 0.00 hours)
5 No errors found.
```

Pg_missing vias

```
Check net GND vias...
Number of missing vias: 0
Checking net GND vias took 1 seconds.
Check net VDD vias...
Number of missing vias: 0
Checking net VDD vias took 0 seconds.
Overall runtime: 1 seconds.
```

pg_connectivity

```
Loading cell instances...
Number of Standard Cells: 24682
Number of Macro Cells: 0
Number of IO Pad Cells: 0
Number of Blocks: 0
Loading P/G wires and vias...
Number of GND Wires: 1029
Number of GND Vias: 1690
Number of GND Terminals: 0
Number of VDD Wires: 1036
Number of VDD Vias: 1702
Number of VDD Terminals: 0
*****Verify net GND connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 22882
Number of floating hard macros: 0
Number of floating I/O pads: 0
Number of floating terminals: 0
Number of floating hierarchical blocks: 0
*****
*****Verify net VDD connectivity*****
Number of floating wires: 0
Number of floating vias: 0
Number of floating std cells: 22882
Number of floating hard macros: 0
Number of floating I/O pads: 0
Number of floating terminals: 0
Number of floating hierarchical blocks: 0
*****
Overall runtime: 0 seconds.
```

Inquiries

Q 1

When defining a maximum routing layer, is it included in for routing or excluded?

Q 2

Is it necessary to match the stub width with the routing track width?

Q 3

Should vias be stacked or not?

Q 4

Should create pg vias for each two consecutive layers or from M2 to M7 at once?

Since creating pg vias from M2 to M7 output is unexpected

```
for {set i 2} {$i < 8} {incr i} {  
    create_pg_vias -nets "GND VDD" -from_layers M$i -to_layers M[expr $i+1]  
}  
  
# M1-M2 via  
create_pg_vias -nets "GND VDD" -from_layers M1 -to_layers M2 -allow_parallel_objects
```

Thank you.